



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)
BSc CHEMISTRY**

Programme	B. Sc. Chemistry				
Course Title	ORGANIC CHEMISTRY V				
Type of Course	MAJOR				
Semester	VII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Preliminary idea about basics of stereochemistry, C-C bond formation, photochemistry, free radical mechanisms, organic reactions and reagents.				
Course Summary	This course explores the basics of stereochemistry, C-C bond formation, photochemistry, free radical mechanisms, organic reactions and reagents.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To understand and apply concepts of stereochemistry, conformations and asymmetric synthesis	U	C	Test /Seminar
CO2	To apply principles of physical organic chemistry including acidity, basicity and reaction mechanisms	U	p	Discussion/ Assignment
CO3	To demonstrate ability to generate carbanions and conduct a variety of condensation and other organic reactions	An	P	Quizzes/Test
CO4	To analyze the role of photochemistry in chemical	Ap	P	Discussion/Seminar /Assignment

	reactions and apply concepts to radical chain reactions			
CO5	To Conduct common organic reactions and apply select reagents in redox and substitution reactions	Ap	P	Assignment/Test
CO6	To empower students in setting up the reaction and purification process	Ap	P	Lab work/Viva
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Stereochemistry III		13	28
	1	Optical isomerism shown by molecules have stereocenters other than C, N, P, S. Axial chirality - Planar chirality and helicity. Nomenclature - Prochirality, Re, Si Nomenclature.	3	
	2	Stereoselective and stereospecific reaction - Effect of configuration on Substitution - Addition and Elimination reactions.	2	
	3	Conformations of 1,2 disubstituted compounds - Carbonyl compounds - Mono and disubstituted cyclohexanes, A-value -Decalin, Bridged bicyclic systems, Anomeric effect, The effect of conformation on reactivity.	4	
	4	Asymmetric synthesis: - Resolution methods - Chiral pool approach - Chiral auxiliary approach - Chiral reagents - Chiral catalyst. (SAMP/RAMP method - Sharpless epoxidation)	4	
II	Physical Organic Chemistry		10	22
	5	Acidity and basicity of organic compounds - Equilibrium - Rate- Rate limiting step - Intermediates and transition states, Reaction profile diagrams, Kinetic and Thermodynamic control of reactions.	5	
	6	Hammond postulate Curtin-Hammett principle.	1	
	7	Methods of determining Reaction Mechanisms	1	
	8	Isotopic labeling, Kinetic isotopic effects, Crossover studies, Detection of intermediates	3	
III	Formation of C-C bonds		13	28
	9	Generation of carbanions from carbonyl compounds, Lithium enolates, Enamines and Silyl enol ethers, O and C-alkylation.	2	

	10	Cram's rule and Felkin Ahn model	2	
	11	Aldol condensation from enolates - Enamine and silyl enol ethers, Mukyamma aldol reaction - Zimmerman Traxler model, Intramolecular reactions.	3	
	12	Claisen condensation - Perkin Reaction - Knoevenagel reaction, Conjugate addition	2	
	13	Robinson annulation - Wittig and related reactions, Reactions of enols - Acid-mediated reactions of aldehydes and ketones.	2	
	14	Organometallic reagents- Grignard reagents - Alkyl lithium agents, Preparation and its reaction with carbonyl compounds and nitriles.	2	
IV	Photochemistry and free radical reactions		9	20
	15	Photochemistry: Fate of an excited molecule - Chemical reactions of excited molecules,	1	
	16	Photochemistry of carbonyl compounds: Norrish type I and II cleavage – α -cleavage, γ -hydrogen abstraction, Paterno Buchi reaction.	2	
	17	Isomerization (cis-trans isomerization in retina, isomerization in benzene), Photosensitization, Di-pi-methane rearrangement, Oxa-di-pi-methane rearrangement.	2	
	18	Free radical reactions: Radical chain reaction, NBS allylic bromination, Acyloin reaction, HLF reaction, Hunsdiecker-Borodin reaction	2	
	19	Generation of C radicals from alkyl halides using AIBN-tributyltin hydride and their cyclizations (5-exo mode only). Radical inhibitors, methods of detecting radical intermediates.	2	
V	Practicals		30	
	1.	Introduction to organic lab	4	
	2	1. Double stage preparations (iodobenzene from aniline/benzil benzilic acid/triphenyl imidazole and its dimerization/ Hydroquinone-benzoquinone anthracene/ caprolactam/ bromoaniline from acetanilide or any reaction based on oxidation/reduction/condensation/rearrangement (purification of the prepared compounds by recrystallization and measurement of melting point) 2. Column chromatography 3. Steam distillation 4. Thin layer chromatography	20	
	3	Open ended	6	

References

1. Organic Chemistry, by Jonathan Clayden, Nick Greeves, Stuart Warren, Oxford University Press
2. March's Advanced Organic Chemistry: Reactions, Mechanisms and Structure, Michael B. Smith, Wiley
3. Organic Chemistry Paperback, Francis Carey, McGraw-Hill Education
4. Advanced Organic Chemistry, David E. Lewis, Oxford Univ Pr; Illustrated edition
5. Principles of Organic Synthesis, R. O. C. Norman & J. M. Coxon, 3rd Ed., CRC press, 2000.
6. Organic Mechanisms- Reactions, Stereochemistry and Synthesis, R. Bruckner, Springer 2007.
7. The Art of Writing Reasonable Reaction Mechanisms, R. B. Grossman, 3rd Ed., Springer, 2019.
8. Modern Physical Organic Chemistry, E. V. Anslyn & D. A. Dougherty, University Science Books, 2015.
9. Name reactions J J Li, Springer.
10. Stereochemistry of Organic Compounds Ernest L. Eliel and Samuel H. Wilen,
11. Stereochemistry of Organic Compounds, D. Nasipuri, 4th Ed., New Age Publications, 2020.
12. Stereochemistry: Conformation and Mechanism, P S Kalsi, New Age International Publishers
13. Vogel's practical organic chemistry
14. Quinone Synthesis and a Visual Introduction to Column Chromatography: An Undergraduate Experiment, Danielle L. Pearson and Russell R. A. Kitson, J. Chem. Educ. 2022, 99, 3731–3734, <https://doi.org/10.1021/acs.jchemed.1c00940>

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	-	-	-	-	-	3		2		2		2
CO 2	2		-	-	-	-	3		2		3		2
CO 3	2	-		-	-	-	3		2		3		3
CO 4	2	-			1	-	3		2		2		3
CO 5	3		-	-	-	-	3		1		2		3
CO 6	-	-	3		-	-	3		2		2		3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment/ viva/seminar	Practical skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2		✓		✓
CO 3	✓			✓
CO 4		✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	✓